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4 August 1975
SUPERSEDING
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(See 6.6)

MILITARY SPECIFICATION

INDICATOR, MULTI-VOLTAGE, MULTI-FREQUENCY AND POLARITY, HAND, PORTABLE

This specification is approved for use by the Naval Sea Systems Command and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers a portable multi-voltage, multi-frequency and polarity indicator. The indicator consists of a hand-held instrument incorporating visible means of recognizing voltage, frequency, and polarity with safe means for energization from electric circuits. The portable indicator is one unit for indicating voltage for direct current (d.c.), voltage and polarity, 60-Hertz (Hz) alternating current (a.c.) voltage and polarity, and 400 Hz a.c. voltage and polarity.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein:

SPECIFICATIONS

FEDERAL

L-P-393 - Plastic Molding Material, Polycarbonate, Injection and Extrusion.

MILITARY

MIL-E-917 - Electric Power, Equipment, Basic Requirements for (Naval Shipboard Use).
MIL-W-13169 - Wire, Electrical (for Instrument Test Leads).
MIL-L-15098 - Lamps, Glow.
MIL-E-17555 - Electronic and Electrical Equipment, Accessories, and Repair Parts, Packaging and Packing of.
MIL-I-45208 - Inspection System Requirements.
MIL-I-46058 - Insulating Compound, Electrical (for Coating Printed Circuit Assemblies).

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-454 - Standard General Requirements for Electronic Equipment.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Qualification. Indicators furnished under this specification shall be products which are qualified for listing on the applicable qualified products list at the time set for opening of bids (see 4.3 and 6.3).

3.2 Material. The case and insulated test prods shall be made of polycarbonate plastic material conforming to L-P-393 or shall be a material having equal, or better, mechanical and electrical characteristics.

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3.3 Design and construction. The indicator shall be so designed and constructed that it can be readily and easily used by one person. The case shall be formed to rest comfortably in the palm of the hand, shall not exceed a maximum weight of 16 ounces and maximum dimensions of 6-1/2 by 3 by 1-1/2 inches. The case shall be provided with anchorages or orifices designed for the insertion of the prods for mechanical protection of the prod points, lead wires and handles during non-use handling and storage. Prods shall fit the anchorages or orificies properly so that they are retained under normal handling but shall be capable of being readily removed for use. Test prod parts shall not become loose or separate. Face plates, when used (see 3.3.2.2), shall be securely attached to the case. The case shall contain a socket in which either prod may be mounted so that the case and the prod may be manipulated as a unit. The lead insulation shall not be damaged or stressed when a prod is inserted in the socket frequently. Signals shall be plainly visible under conditions of shadow, and shall be clearly visible at all angles from perpendicular to the indicator face to 45 degrees from the perpendicular. The visual signal shall have a light output of at least 2 foot lamberts. Appurtenances shall not protrude to prevent snagging of adjacent objects. The indicator shall be compact, sufficiently rugged to be carried in a tool kit, and shall be capable of accomplishing the functions, regardless of the position in which it is held. The following requirements of MIL-STD-454 shall apply:

Requirement 1 - Safety
Requirement 5 - Soldering
Requirement 9 - Workmanship (personnel Hazard)
Requirement 17 - Printed wiring boards (if used). Part identification by reference designation is not required.
Requirement 23 - Adhesives
Requirement 30 - Semiconductor devices

When printed wiring boards are used they shall have a conformal coating conforming to MIL-I-46058 or shall be a clear coating which is easily removable with a soldering iron without damage to the printed wiring board. Electrical creepage and clearance distance requirements of MIL-E-917 shall apply. Adhesive joints shall be securely made and shall maintain integrity under the environmental conditions specified herein. The drop test of 4.8.12 shall not cause mechanical damage to indicator and shall not affect the electrical performance. The indicator shall show no signs of internal damage or over heating (see 6.5.1) following application of 600 volts (V) root-mean-square (rms) 60 Hz for one minute (see 4.8.11).

3.3.1 Indicator voltage response. The indicator shall provide a continuous response to voltage within the range of 25 to 600 Vd.c. or Va.c. (60 Hz and 400 Hz). A minimum of two readouts shall respond to provide simultaneous fail-safe indications of the presence of a.c. or d.c. power (for example, readouts shall respond to Vd.c. and polarity or Va.c. and frequency). Below 90V however, the two simultaneous readouts may be voltage indications.

3.3.1.1 Polarity response. For Vd.c., the indicator shall also provide independent indications and identification of positive and negative terminals. Polarity indications will not be required for voltages less than 90V. Polarity indications shall not respond to a.c. inputs.

3.3.1.2 Frequency response. For Va.c., the indicator shall also provide independent indications and identification of 60 Hz and 400 Hz power sources. The frequency indicators shall respond to variations of plus and minus 10 percent of the nominal power frequencies.

3.3.2 Readout indications. The indicator shall provide discrete light emitting readouts for voltage, polarity and frequency as specified in 3.3.4. Readouts may be constant glow or flashing indicators. Flashing indicators shall have a minimum frequency of 70 flashes per minute over the voltage range of 25 to 600 V. Direct readings shall be provided and shall not require manual adjustment or calculation to interpret.

3.3.2.1 Voltage readouts. Voltage readouts shall be provided to respond to 28, 115, 220 and 440 Va.c. or to 28, 115, 250 and 500 Vd.c. nominal voltages. A readout indicator shall respond to voltages within 10 percent of the given nominal values. For voltages outside the 10 percent limit, either the higher or lower nominal readout indicator shall respond.

3.3.2.2 Readout identifications. Markings shall be provided to identify the specified nominal Va.c. and Vd.c. readouts specified in 3.3.2.1, the frequency readouts specified in 3.3.1.2 and the polarity readout specified in 3.3.1.1. The markings shall be indelibly applied on a face plate or on the indicator case. The markings shall be legible and be unaffected by normal handling or rubbing with petroleum products, water, perspiration or other typical shipboard materials. The lettering shall exhibit sharp contrast to the background surface and be readable from a distance of 28 inches with an illumination level of 7 foot candles.

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3.3.3 Leads and terminals. Two flexible insulated test leads, not less than 24 inches in length shall be furnished as integral parts of the indicator. Test leads shall be one colored black and one colored red and shall conform to MIL-W-13169. Each lead shall terminate at one end in a sharp pointed test prod with an insulated handle (see 3.2) provided with an enlarged shoulder adjacent to the bare end to limit finger slippage. The connection between the lead wire and the prod tip shall be a secure mechanical connection or a soldered joint which is mechanically secure before soldering. The test prods shall consist of a completely insulated section at least 3-1/2 inches long with a pointed electrically conducting tip about 1/2 inch long which is connected to the lead wire. The lead shall be so secured within the case and prod barrel that no external strain on the leads will be communicated to the internal connections within the indicator or prod. One of the insulated handles shall be red and the other black. The handles shall be designed to fit securely the prod storage orifices supplied.

3.3.4 Indicating devices. Glow lamps, when used, shall meet the requirements of MIL-L-15098. The indicating device shall be secured in such a position that the minimum voltage indication is readily visible throughout the maximum viewing angle of the device. The indicating device shall not be dislocated by normal handling or by the drop test specified in 4.8.12.

3.4 Heat effect at 65 degrees Celsius (°C). Indicators shall be capable of withstanding the heat effect test at 65°C without causing performance to deviate from 3.3.1 and 3.3.2 (see 4.8.5).

3.5 Low temperature exposure. Indicators shall be capable of withstanding the low temperature exposure of 0°C for 16 hours without causing performance to deviate from 3.3.1 and 3.3.2 (see 4.8.6).

3.6 Dielectric withstanding voltage. Indicators shall operate normally and shall show no internal damage of flashover when tested as specified in 4.8.7.

3.7 Insulation resistance. The insulation resistance of the indicators shall be not less than 20 megohms when corrected to 25°C (see 4.8.8).

3.8 Humidity. Indicators shall withstand the humidity test of 4.8.9 without damage to any parts and without interference to the operation of the tester.

3.9 Temperature rise. Temperature rise shall not exceed 25°C at any area on the exterior of the case, and parts shall not overheat (see 6.5.1), when subjected to the test of 4.8.10.

3.10 Life. The indicators shall meet the requirements of this specification when subjected to the test of 4.8.13 (see 4.3.2).

3.11 Identification and marking. The indicators shall be permanently and legibly marked with the following minimum information:

(a) Manufacturer's name or trade-mark, or both.
(b) Manufacturer's type or model number, or both.
(c) National stock number.

3.12 Operating and maintenance instructions. An operating and maintenance instruction sheet shall be provided with each indicator inside an individual wrapping or otherwise temporarily fastened to each unit. The instruction sheet shall consist of simple operating instructions, description of capabilities, safety precautions, and care of the indicator (see 4.3.1).

3.13 Workmanship. Workmanship shall be in accordance with requirement 9 of MIL-STD-454. Test lead insulation shall show no signs of cracking, abrasion or deterioration.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

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4.1.1 Inspection system. The supplier shall provide and maintain an inspection system acceptable to the Goverment for supplies and services covered by this specification. The inspection system shall be in accordance with MIL-I-45208.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

(a) Qualification inspection (see 4.3).
(b) Quality conformance inspection (see 4.5).

4.3 Qualification tests. Qualification tests shall be conducted at a laboratory satisfactory to the Naval Ship Engineering Center. Qualification tests shall consist of the tests specified in table I. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3 and 6.3.1).

Table I - Qualification tests.

Examination and tests	Requirement paragraph	Test paragraph
General examination	3.2, 3.3, 3.11, and 3.13	4.6.1
Creepage and clearance	3.3	4.8.1
Position influence	3.3	4.8.2
Indication	3.3.1 and 3.3.2	4.8.3
Voltage and frequency variation	3.3.1 and 3.3.2	4.8.4
Heat effect at 65°C	3.4	4.8.5
Low temperature exposure	3.5	4.8.6
Humidity	3.8	4.8.9
Temperature rise	3.9	4.8.10
Rated voltage	3.3	4.8.11
Drop test	3.3 and 3.3.4	4.8.12
Insulation resistance	3.7	4.8.8
Dielectric withstanding voltage	3.6	4.8.7
Life test	3.10	4.8.13
Final test	-----	4.8.14
Final examination	-----	4.6.2

4.3.1 Included with the qualification inspection report (see 4.3), a detailed report showing that the lead wire insulation conforms to MIL-W-13169 (see 3.3.3) shall be submitted by the wire supplier, and also the operating and maintenance instruction sheet specified in 3.12.

4.3.2 Qualification samples. Three sample indicators shall be used for qualification testing. One sample shall be subjected to the tests specified in table I in the order shown. The remaining two samples shall be subjected to the following tests in the order shown:

(a) Indication (see 4.8.3).
(b) Life test (see 4.8.13).
(c) Final test (see 4.8.14).

4.4 Sampling for quality conformance inspection.

4.4.1 Lot. For purposes of sampling, a lot shall consist of all indicators offered for delivery at one time.

4.4.2 Sampling for general examination. A random sample of indicators shall be selected from each lot in accordance with MIL-STD-105 at inspection level II for the examination of 4.6.1. The AQL shall be 1.0 percent defective for major defects and 4.0 percent defective for minor defects. For major and minor defects, if the number of defective indicators in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected. Any critical defect in any indicator will be cause for rejection of the lot represented by the sample.

4.4.3 Sampling for group B inspection. A random sample of indicators shall be selected from each lot in accordance with MIL-STD-105 at inspection level II with an AQL of 1.0 percent defective, and shall be subjected to the group B inspection of 4.5.2.

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4.5 Quality conformance inspection. Quality conformance inspection shall consist of the group A and B inspection of 4.5.1 and 4.5.2.

4.5.1 Group A tests. Each indicator shall be subjected to the rated voltage test of 4.8.11 and the results shall be compared with the requirements of this specification. Failure to conform to the test of 4.8.11 shall be counted as a defect. Defective units shall not be offered for delivery.

4.5.2 Group B inspection. Sample indicators selected in accordance with 4.4.3 shall be subjected to each of the group B tests shown in table II in the order shown, and the results shall be compared with the requirements of this specification. Failure to conform to the requirements of this specification for any group B test shall be counted as a defect and the defective indicator shall be rejected. If the number of such nonconforming indicators in any sample exceeds the acceptance number for that sample, the lot represented by the sample shall be rejected.

Table II - Group B inspection.

Examination and tests	Requirement paragraph	Test paragraph
Indications	3.3.1 and 3.3.2	4.8.3
Drop test	3.3 and 3.3.4	4.8.12
Insulation resistance	3.7	4.8.8
Dielectric withstand voltage	3.6	4.8.7
Final test	-----	4.8.14
Final examination	-----	4.6.2

4.5.3 Rejected lots. If an inspection lot is rejected, the supplier may rework the lot to correct defects, or screen out defective units and again inspect the lot. Such lots shall be sampled in accordance with tightened inspection of MIL-STD-105.

4.6 Examination.

4.6.1 General examination. Each sample indicator selected in accordance with 4.4.2 shall be subjected to a general examination to ascertain that the material, workmanship and design are in accordance with this specification. Defects shall be classified as shown in table III.

Table III - Classification of defects.

Critical	Defects
1	Indicator does not indicate or indications are not as specified (see 3.3.1 through 3.3.2.2).
2	Signals not plainly visible under conditions of shadow and at all angles from perpendicular to indicator face to 45 degrees from perpendicular.
3	Does not accomplish function in any position held.
4	Test lead insulation shows signs of cracking, abrasion, or deterioration.
Major	**Defects**
101	Molded plastic parts are not as specified.
102	Prod storage orifices are not provided or do not protect prod points, lead wires, and handles.
103	Prod socket is not provided.
104	Prods not capable of being readily removed.
105	Test prod points loose or separated.
106	Handles do not fit securely in prod storage orifices.
107	Indicating devices are not as specified or not correctly positioned.
108	Face plate markings not as specified.
109	Reading not direct, requires manual adjustment, or mental calculation.
110	Face plate is not securely attached to case.
111	Printed wiring boards, adhesives, semiconductor and conformal coatings are not as specified.
112	Width, overall exceeds 3 inches.
113	Length, overall exceeds 6-1/2 inches.
114	Depth, overall, exceeds 1-1/2 inches.
115	Weight, overall, exceeds 16 ounces.
116	Instruction sheet missing or does not conform to requirements of 3.12.
117	Leads and terminals not in accordance with 3.3.3.

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Table III - Classification of defects (con.).

Minor	Defects
201	Indicator will not rest comfortably in palm of hand.
202	Color of handles not as specified
203	Identification marking not as specified.
204	Indicator has protrusions that could snag adjacent objects.
205	Workmanship (except those items listed separately in this table).

4.6.2 Final examination. After completion of all tests, a final examination shall be made. All adhesive joints and labels shall be examined gently with a knife or other instrument to determine if separation, lifting of corners, or curling of edges has started. Test lead insulation shall be examined for signs of cracking, abrasion or deterioration. Test prods shall be examined for signs of looseness, separation, or improper fit within anchorages or orifices.

4.7 Test conditions. All tests shall be conducted at an ambient temperature between 20° and 35°C, unless otherwise specified herein.

4.8 Test procedures.

4.8.1 Creepage and clearance. Creepage and clearance distances shall be demonstrated by actual measurement.

4.8.2 Position influence. Voltage indicators shall be tested to determine conformance to 3.3, while positioned such that each of the six sides of the indicator, in turn, is the lower horizontal surface of the indicator.

4.8.3 Indication test. The indicator shall be energized, the voltage monitored by a standard voltmeter and the indication of voltages, frequencies and polarities as specified in 3.3.1 through 3.3.2.2 confirmed at 60 and 400 Hz and for d.c. with prods at both polarities.

4.8.4 Voltage and frequency variations. The indicator shall be tested at variations of plus or minus 10 percent of nominal voltages (28, 115, 220, 440 V a.c.; 28, 115, 250 and 500 V d.c.) with applicable concurrent variations of plus or minus 10 percent of nominal frequencies (60- and 400-Hz) (see 3.3.1 and 3.3.2). The indication shall be tested also by application of a voltage that is varied continuously from 25 to 600 V at both 60 and 400 Hz and for d.c. with prods at both polarities to determine that at least two indications are given of power to the prods at all voltages between 25 and 600 V.

4.8.5 Heat effect at 65°C. Heat effect shall be determined as follows: Place indicator in chamber at 65°C. Maintain that temperature for 16 hours. Allow the indicator to cool to 25°C over a period of 8 hours. Repeat this cycle three times. Performance of the indicator shall be checked during the final test of 4.8.14.

4.8.6 Low temperature exposure. Indicators shall be exposed to a temperature of 0° ± 3°C for 16 hours and at the end of the period voltage readings shall be taken as specified in 4.8.3. Indicators shall then be maintained at room temperature for sufficient time to attain thermal stability. Performance of the indicator shall be checked during the final test of 4.8.14.

4.8.7 Dielectric withstanding voltage. For the dielectric withstanding voltage test, indicators shall be encircled by a metallic ribbon. Indicators shall be subjected to 1900 V rms or the d.c. value (3230 V) applied between the prod points electrically connected together and the following:

(a) The metallic ribbon specified herein.
(b) All other exposed metal parts.

4.8.7.1 The voltage shall be nominal 60 Hz in frequency, and shall approximate, as closely as possible a true sine wave in form. The test voltage shall be applied continuously for a period of 60 second(s). The test voltage shall be measured by the voltmeter method whereby the measuring meter derives its voltage from the high-potential circuit, either directly or by means of a voltmeter coil placed in the testing transformer, or by means of an auxiliary ratio transformer. To avoid voltage surges the applied voltage shall be raised to other full value gradually, and held for 60 s. Upon completion of the 60 s test period, the voltage shall be gradually reduced. Dielectric breakdown between the prods and metallic ribbon, or other exposed metal parts, will be considered a failure to conform to this test.

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4.8.8 Insulation resistance. For the insulation resistance test, indicators shall be encircled by a metallic ribbon. Using a 500-V, d.c. insulation test, the insulation resistance shall be determined between the prod points electrically connected together and the following:

(a) The metallic ribbon specified herein.
(b) All other exposed metal parts.

4.8.9 Humidity. The indicator shall be placed in a dry oven at a temperature of 40° ± 5°C for a period of 24 hours. The indicators shall then be placed in a chamber and subjected to a relative humidity of 95 ± 5 percent and a temperature of 40° ± 5°C for a period of 100 hours. Upon completion of this period, indicators shall be permitted to dry at room temperature for a period of 24 hours.

4.8.10 Temperature rise. The indicator shall be continuously energized at 440 V, 60 Hz for 1/2 hour. Temperature rise readings of the exterior of the case shall be taken at the warmest area on the case. This test shall be repeated at 440 V, 400 Hz and at 250 V d.c. Between each test, the unit shall be allowed to cool to room temperature.

4.8.11 Rated voltage test. With the case open, a voltage of 600 V rms 60 Hz shall be applied across the indicator prod points for 1 minute (see 3.3.3). Internal parts shall be observed for overheating during this test. The indicator shall then be subjected to a visual examination for damages before the indicator case is closed.

4.8.12 Drop. A drop test shall be made to determine conformance to the rugged construction requirements of 3.3. The drop test shall consist of dropping the indicator with leads and prods free and at the same level from a height of 3 feet to a flat concrete floor. There shall be no mechanical damage to the indicator, leads or prods which expose internal conductors or affects the use of the unit for the designed purpose. The indicating device shall not be dislocated during the drop test (see 3.3.4).

4.8.13 Life test. The life test shall consist of cycles of impressing a series of voltages to the prods of the indicators for 2 s each ±1/2 s and deenergized for 10 to 15 s after each application of voltage. Cycling shall be continued for 8 hours per day for a period of 30 days. The impressed voltages shall consist of each voltage (±5 percent) at each frequency and d.c. specified in 3.3.2.1. Half of the d.c. voltages shall be of one polarity and half shall be of the opposite polarity. The time of exposure to each specific signal shall be approximately equal. After each 48 hours, the indicator shall be checked to be sure all indications are activated.

4.8.14 Final test. After completion of all of the other tests, the indication test of 4.8.3 shall be performed to determine conformance with 3.3.2.

4.9 Inspection of preparation for delivery. The packaging, packing, and marking shall be inspected for compliance with section 5 of this specification.

5. PREPARATION FOR DELIVERY

(The preparation for delivery requirements specified herein apply only for direct Government procurements. For the extent of applicability of the preparation for delivery requirements of referenced documents listed in section 2, see 6.4).

5.1 Preservation and packaging, packing, and marking. The equipment and accessories, repair parts, and technical publications shall be preserved and packaged level A or C packed level A or B, or C, as specified (see 6.2.1) and marked in accordance with MIL-E-17555.

6. NOTES

6.1 Intended use. The indicators covered by this specification are intended for use by an electrician in normal maintenance operations to check nominal line voltages in power and lighting circuits or under casualty conditions by damage control personnel to identify unknown circuits. It will serve as a Safety device to inform the electrician or damage control personnel if a circuit is de-energized and safe to work on or if it is energized and requires further action to ensure personnel safety.

6.2 Ordering data.

6.2.1 Procurement requirements. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Level of preservation and packaging required (see 5.1).

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6.3 With respect to products requiring qualification, awards will be made only for products which are at the time set for opening of bids, qualified for inclusion in applicable Qualified Products List QPL 23830 whether or not such products have actually been so listed by that date. The attention of the suppliers is called to this requirement, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. The activity responsible for the Qualified Products List is the Naval Ship Engineering Center, Prince George's Center, Center Building, Hyattsville, Maryland 20782, and information pertaining to qualification of products may be obtained from that activity. Application for Qualification tests shall be made in accordance with "Provisions Governing Qualification SD-6" (see 6.3.1).

6.3.1 Copies of "Provisions Governing Qualification SD-6" may be obtained upon application to Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, Pennsylvania 19120.

6.4 Sub-contracted material and parts. The preparation for delivery requirements of referenced documents listed in section 2 do not apply when material and parts are procured by the supplier for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.5 Definition.

6.5.1 Overheating. Overheating is defined as exceeding the maximum operating temperature specified by the manufacturer of the part.

6.6 THE MARGINS OF THIS SPECIFICATION ARE MARKED "#" TO INDICATE WHERE CHANGES (ADDITIONS, MODIFICATIONS, CORRECTIONS, DELETIONS) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS WAS DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

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